# Forming Barium Stars through Stable Mass Transfer from TP-AGB Stars

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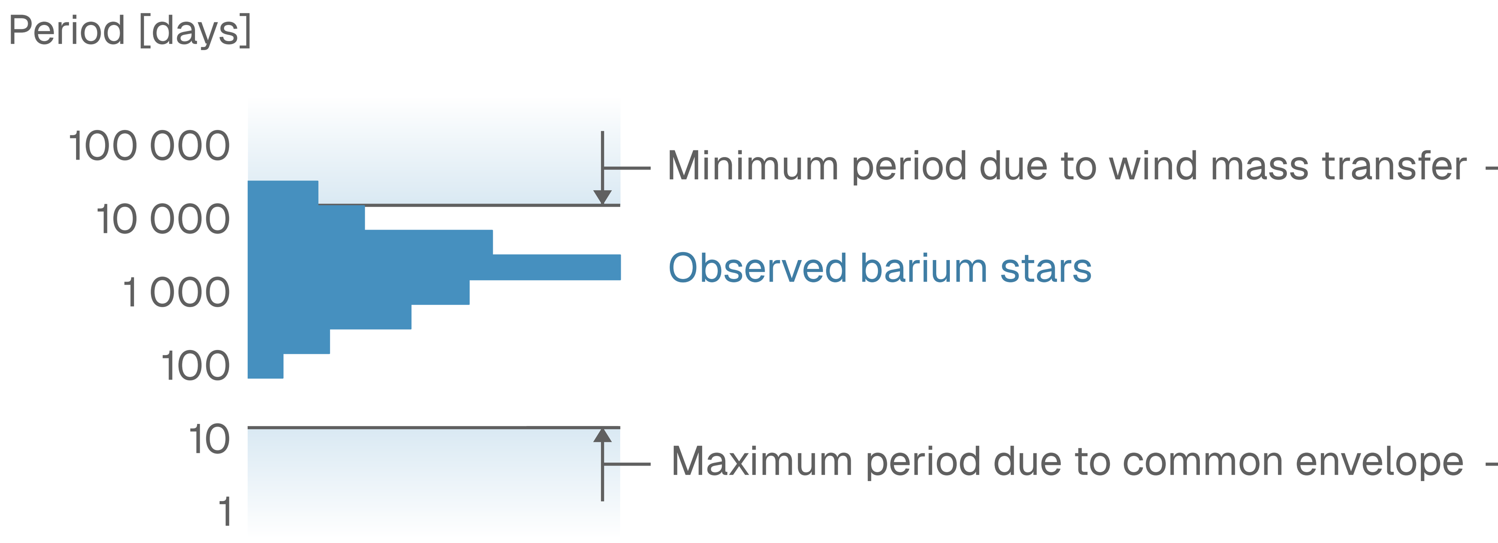
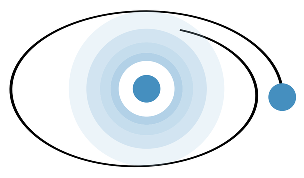
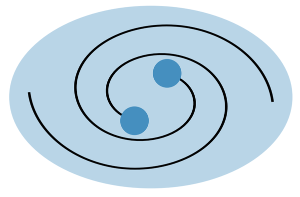
Barium stars are stars enriched in barium by mass accretion from a thermally pulsing (TP) asymptotic giant branch (AGB) star. Barium stars are often observed with periods between 100 and 20 000 days.[1,2]



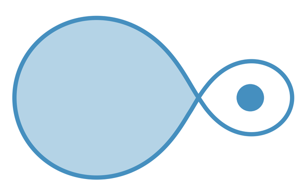

The period distribution of barium stars is difficult to explain:



**Wind mass transfer**
cannot explain the small orbits.



**Common Envelope Events**
result in orbits that are too small.



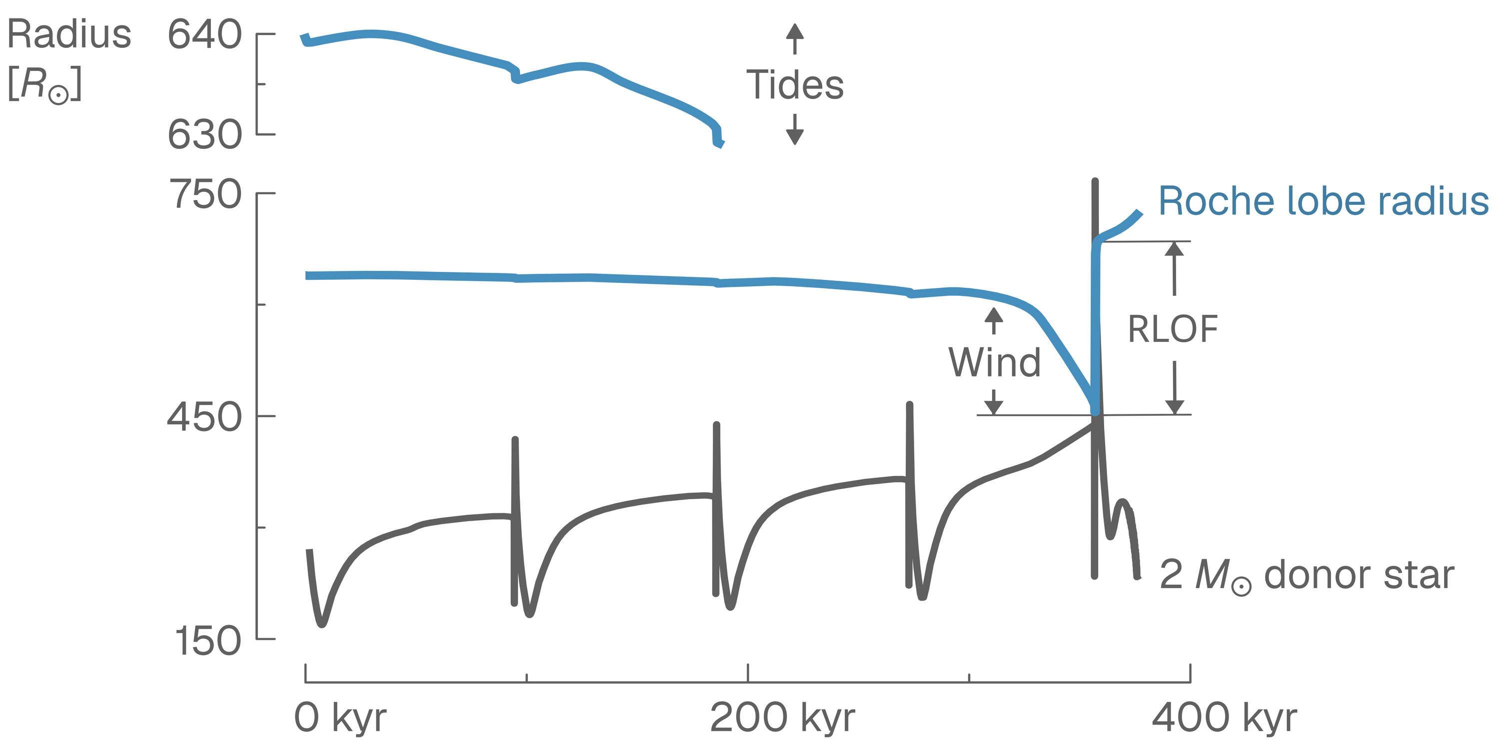
**Roche lobe overflow (RLOF) mass transfer**
is often assumed unstable for giants, *but* recent models[3] show it can be stable in RGB & E-AGB stars.

We attempt to undestand how barium stars form, by exploring if TP-AGB RLOF can explain the properties of observed barium stars.

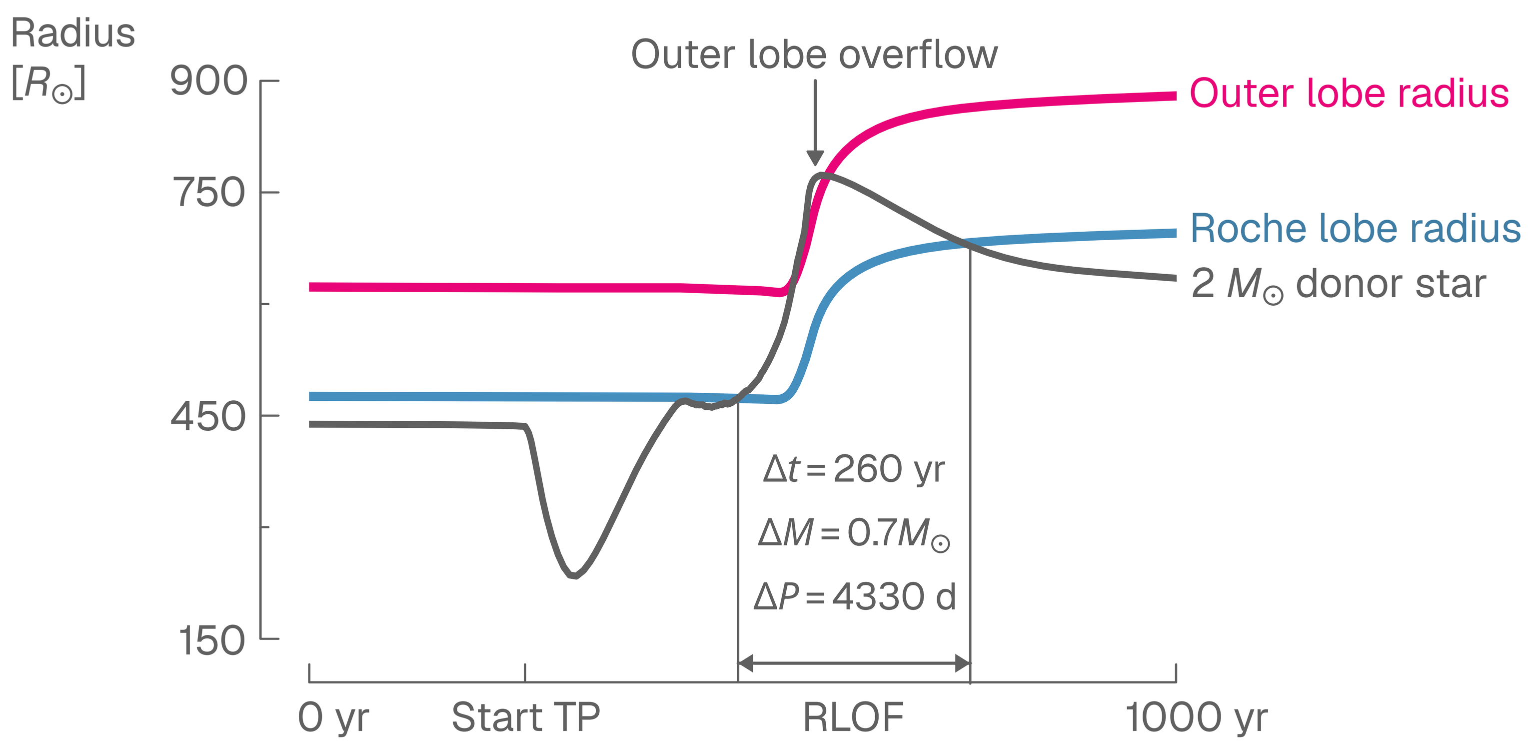
We simulate conservative RLOF from TP-AGB stars using *MESA*, and include dredge-up[4], wind mass transfer[5], and tides[6].

We evaluate the stability of RLOF by analyzing the mass loss rate and the expansion of the donor star. We then determine the orbital effects.

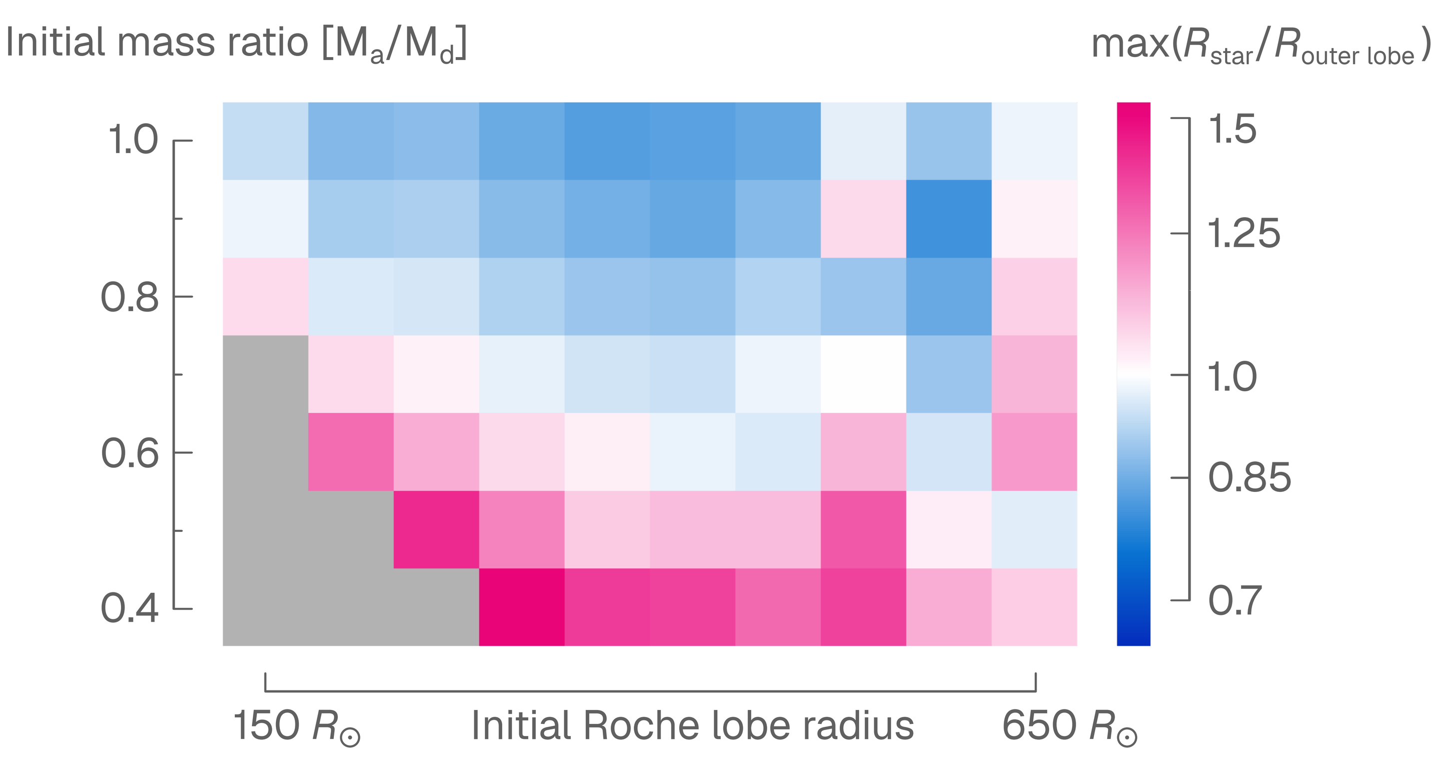
Stellar evolution and winds drive systems into RLOF.



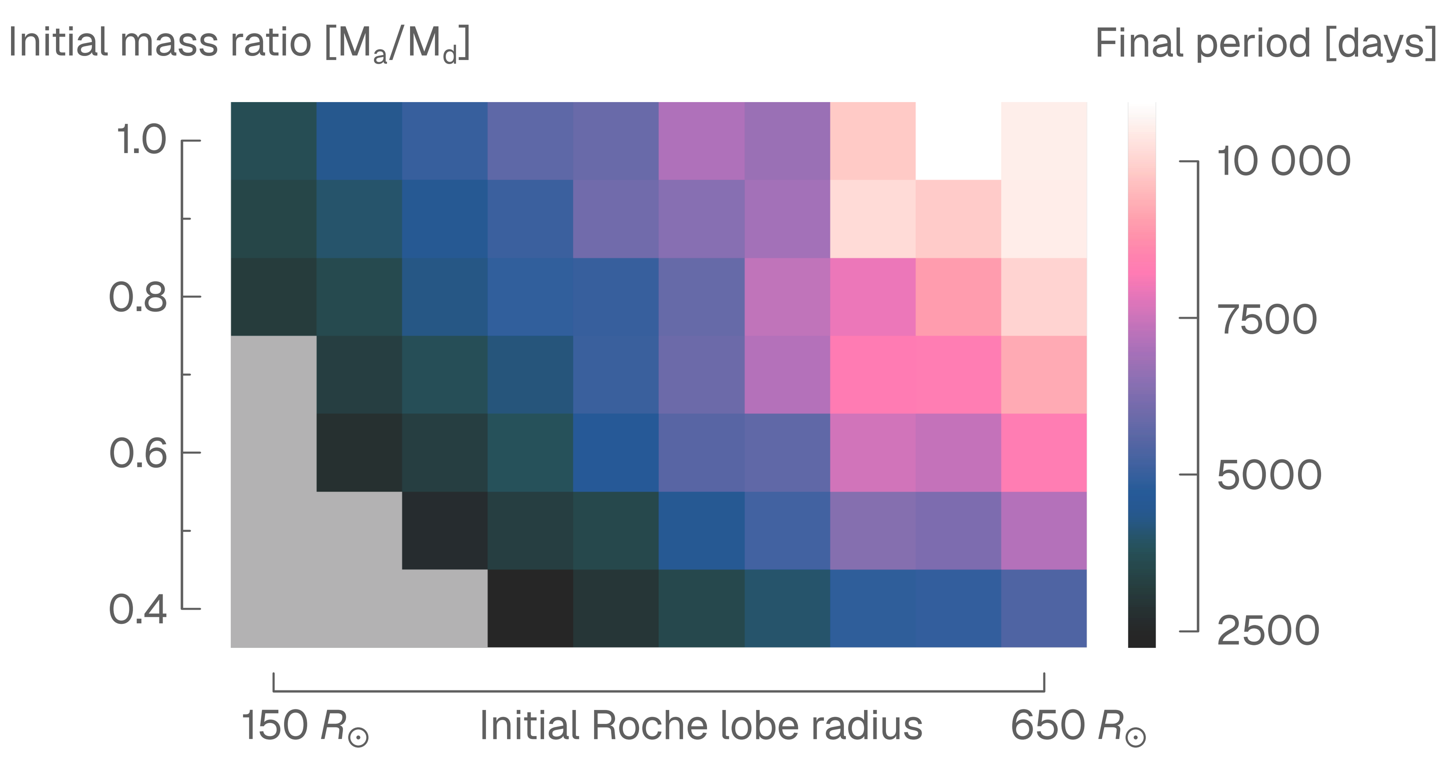

RLOF occurs during the short thermal pulses of the TP-AGB star and results in significant mass transfer and widening of the orbit.




A significant portion of our models overflow their outer Roche lobe. *MESA* is not made to operate in this regime.




Fully conservative RLOF mass transfer results in periods larger than those of the observed barium star population.




In the future, we want to:

- Explore non-conservative mass transfer and mass loss from outer Lagrange points.
- Investigate mass transfer in systems with non-zero eccentricity.
- Thoroughly compare our results from stable mass transfer to properties of barium stars.

[1]: Jorissen, A., Boffin, H. M. J., Karinkuzhi, D., et al. (2019) A&A626, A127.
[2]: Escorza, A. & De Rosa, R. J. (2023) A&A671, A97.
[3]: Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) A&A669, A45.
[4]: Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) MNRAS527, 9643.
[5]: Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & PortegiesZwart, S. (2018) A&A618, A50
[6]: Preece, H. P., Hamers, A. S., Neunteufel, P. G., Schaefer, A. L., & Tout, C. A.(2022) ApJ933, 25.